

SPAM MESSAGE DETECTION USING MACHINE LEARNING

ABSTRACT

This project aims to classify SMS messages as Spam or Ham using Machine Learning techniques. The SMS Spam Collection dataset was used for training and testing. Text messages were converted into numerical features using CountVectorizer. Two machine learning algorithms, Multinomial Naive Bayes and Logistic Regression, were implemented and compared. The model performance was evaluated using Accuracy, Precision, Recall, F1-Score, Cross Validation, and ROC-AUC metrics.

OBJECTIVE

To build and evaluate a supervised machine learning classification model capable of identifying spam messages.

TOOLS AND TECHNOLOGIES

- Python
- Pandas
- Scikit-Learn
- Matplotlib
- VS Code

ALGORITHMS USED

1. Multinomial Naive Bayes
2. Logistic Regression

WORKFLOW

1. Load and preprocess the dataset.
2. Convert text messages into numerical vectors using CountVectorizer.
3. Split the dataset into training and testing sets.
4. Train Naive Bayes and Logistic Regression models.
5. Evaluate performance using Accuracy, Precision, Recall, F1-Score, Cross Validation, and ROC-AUC.
6. Compare the performance of both algorithms.
7. Predict whether a user-entered message is Spam or Ham.
8. Visualize message distribution using a pie chart.

RESULTS

Naive Bayes Accuracy: 97.85%

Logistic Regression Accuracy: 97.76%

ROC-AUC Score: 0.9792

Average Cross Validation Score: 0.9808

The Naive Bayes model slightly outperformed Logistic Regression and achieved excellent classification performance.

CONCLUSION

The Spam Message Detection System successfully classifies SMS messages as Spam or Ham with high accuracy. The project demonstrates text preprocessing, supervised machine learning, model evaluation, algorithm comparison, and visualization techniques.

OUTPUT SCREENSHOTS

```
ROC-AUC Score: 0.9792158894645941
```

```
Cross Validation Scores:
```

```
[0.98295964 0.98206278 0.97935368 0.98114901 0.97845601]
```

```
Average CV Score: 0.980796225777105
```

```
===== LOGISTIC REGRESSION =====
```

```
Accuracy: 0.9775784753363229
```

```
===== ALGORITHM COMPARISON =====
```

```
Naive Bayes Accuracy: 0.97847533632287
```

```
Logistic Regression Accuracy: 0.9775784753363229
```

```
Enter a message: Congratulations! You won a free iPhone. Claim now.
```

```
Result: Spam Message
```

Spam vs Ham Messages

